ASSADUZZAMAN MUNNA

AI/ML & Software Engineer — Aspiring AI/ML Specialist

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EDUCATION

North South University (NSU) — Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering (Department of ECE) 2022 – May 2026
CGPA: 3.74 / 4.00

EXPERIENCE

AI Engineer
The Data Island — April 2026 – Present
- Promoted from Junior AI Intern to engineer and optimize scalable data pipelines, seamlessly integrating enterprise machine learning frameworks for production systems.

Undergraduate Teaching Assistant (UGA)
North South University (ECE Dept.) — June 2025 – Present
- Mentor undergraduate students by facilitating technical sessions, supporting curriculum delivery, and evaluating engineering assignments.

On-the-Job Training (AI Engineering)
Nippon AI Dojo 2025 (Chowa Giken & AI Samurai Japan) — Sep 2025 – Jan 2026
- Developed practical AI implementation and model optimization skills through hands-on, real-world projects mentored directly by Japanese industry experts.

SKILLS

- **AI, ML & Computer Vision:** PyTorch, TensorFlow, OpenCV, Hugging Face, Knowledge Distillation, Vision Transformers, Multimodal AI, MLOps.
- **Generative AI & NLP:** Large Language Models (LLMs), RAG, LangChain, Vector Databases, Fine-tuning, NLP, SAM.
- **Cloud, DevOps & Data:** Google Cloud Platform (GCP), NVIDIA DeepStream, Docker, BigQuery, SQL, Pandas, NumPy, Linux, FFmpeg.
- **Software Engineering & Languages:** Python, C++, JavaScript, Dart, React.js, Next.js, Flutter, RESTful APIs, System Design, Git.

FEATURED PROJECTS

Real-Time Spatial-Temporal Anomaly Detection (Confidential) — *AI Engineer*
- Built and deployed highly optimized **NVIDIA DeepStream** pipelines for critical industrial safety applications, including low-latency fire and smoke detection.

Enterprise Video Management System (VMS) — *System Architect*
- Engineered an automated IP camera discovery and spatial mapping infrastructure to centralize large-scale industrial surveillance networks.

Industrial Automation System (Foil Stamping) — *System Architect*
- Built a low-latency **Python/OpenCV** backend capable of processing high-throughput inspection streams for industrial manufacturing lines.

Agentic Codebase Q&A System — *Personal Project*
- Developed an intelligent agentic architecture utilizing **Retrieval-Augmented Generation (RAG)** and LLMs to automate semantic search and Q&A across complex software codebases.

PUBLICATIONS

SPECTRA: Explainable Shared–Private Multimodal Alignment for Joint Emotion Recognition and Continuous Sentiment Prediction
Moontaha Rawshan, Mohammad Ishzaz Asif Rafid, Al-Amin Rabbi, **Assaduzzaman Munna**, Riasat Khan. — *Submitted to Alexandria Engineering Journal*
- Engineered a shared-private multimodal framework using context-aware cross-attention, improving emotion classification to **75.86%** and sentiment accuracy to **88.33%** across 4 benchmark datasets.

Bone Fracture Detection Using Vision Transformers: A Comparative Analysis of the PiT and CaFormer Models
Atikul Islam Munna, Md. Ibrahim Khalil, **Assaduzzaman Munna**, et al. — *2026 IEEE 2nd Int. Conf. on Secure IoT (SATC)*, 2026. **DOI:** 10.1109/SATC69565.2026.11542322
- Evaluated Vision Transformers against CNNs on 4,083 X-ray images, with the PiT model achieving 97.51% testing accuracy in automated fracture diagnosis.

Hybrid CNN–MobileViT Model with Knowledge Distillation for Efficient Renal Calculi Detection
Assaduzzaman Munna, Mushfika Hossain, Anonto Bormon, Riasat Khan. — *Submitted to SPICSCON 2026. (Under Review)*

- Engineered a knowledge distillation model reducing parameters by **18.5x** for edge deployment, achieving **99.80% test accuracy** with highly optimized inference speeds on an NVIDIA T4 GPU.

ACHIEVEMENTS & CERTIFICATIONS

Nippon AI Dojo 2025 (AI Training Program) — *Selected Participant*
Completed a rigorous program designed to develop practical AI engineering skills via structured lectures and hands-on projects by Chowa Giken and AI Samurai Japan.